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NFT Copymint Detection

A Routed, ORB-Augmented Detector

Final Project Report

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Abstract

NFT copymints reuse the visual content of existing non-fungible tokens while applying image transformations intended to evade duplicate detection. This project extends the perceptual-hash detector studied by Kotzer et al., whose 2-Minimal rule combines **aHash**, **pHash**, **hsvHash**, and **sHash** and reports a duplicate when at least two signals agree. We first reproduced the original hashing pipeline, including a from-scratch C11 implementation used to validate the reference behavior. That reproduction exposed a structural issue: the variable-length **sHash** distance is directional, is not symmetric, and violates the triangle inequality. Consequently, it does not satisfy the metric assumptions required for sound BK-tree pruning; a valid duplicate can be stored in a branch that the search prunes.

The project therefore pivoted from search-speed optimization to detection accuracy. We replaced **sHash** with ORB local feature matching and evaluated the replacement pairwise, after retiring an earlier ORB baseline whose gallery silently excluded all native-resolution CryptoPunks. On the geometric subset, ORB achieved 90.6% F1 compared with an oracle-best 76.4% for **sHash**; at equal precision, recall was 90.0% versus 27.2%. In the full four-signal detector, the replacement improved F1 from 81.7% to 87.5% at a common false-positive operating point. The gain persisted under two-manipulation stress testing (+8.7 F1), but ORB introduced a deployment cost: approximately 14.5 KB of descriptors per image at the default setting, far beyond the compact on-chain metadata model of the original detector.

We also built a query-only manipulation-aware router using 93 absolute image features and a Random Forest. Although the router predicted the relevant reliability states accurately within the generated distribution, dynamic routing added essentially zero end-to-end gain. Experiments showed why: broken similarity signals usually become silent rather than noisy, so dropping one rarely changes a two-vote quorum. Multi-manipulation tests then showed that the two-failure state needed for useful quorum relaxation is difficult to observe with the tested cues: severe pixelation, the main detail-destroying transformation, also erases the color forensic cue used to identify a simultaneous color change. The project therefore yields both a positive result - the **sHash**-to-ORB replacement - and a negative but mechanistically explained result: routing does not measurably improve the detector under the tested manipulation model.

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1 Introduction

Non-fungible tokens (NFTs) are blockchain-recorded assets that may refer to digital artwork, images, or other media. The blockchain can establish token ownership and transaction history, but it does not prevent a second party from copying an existing image, modifying it slightly, and minting a visually similar token. Such copies are commonly referred to as *copymints*. Their detection is therefore an image-similarity problem rather than a cryptographic-integrity problem.

Cryptographic hashes are intentionally unsuitable for this task: changing a single byte should produce an unrelated digest. Perceptual hashes have the opposite goal. They map visually related images to compact representations for which a small distance indicates similarity. The detector that motivated this project combines four such signals - **aHash**, **pHash**, **hsvHash**, and **sHash**- and reports a duplicate when at least two of the four fall within configured thresholds. The individual hashes are meant to be complementary: one signal may fail after rotation, another after a color edit, while agreement between two independent signals suppresses false positives.

Our project began as an attempt to improve this detector without changing its answer, primarily by routing queries only to useful signals. The direction changed when reproduction work revealed that one of the four representations, **sHash**, does not satisfy the mathematical assumptions of the BK-tree used to search it. This was not a small implementation detail: if the distance is non-metric, pruning can remove branches that contain valid neighbors. At that point, optimizing the existing four-tree design was less meaningful than first correcting the underlying representation and evaluation pipeline.

The project therefore evolved around four questions. First, can the original detector be reproduced closely enough that its behavior can be investigated with confidence? Second, is **sHash** compatible with the metric search structure used for it? Third, if **sHash** is replaced, can a stronger signal close the spatial and geometric blind spots without simply moving the problem elsewhere? Fourth, can a query-only router learn when particular signals are trustworthy and use that knowledge to improve the final detector?

The answers are mixed, which is precisely what makes the project useful. ORB local feature matching is a much stronger spatial-structure signal than **sHash** in our evaluation, and the full detector improves when ORB replaces **sHash**. However, ORB emits hundreds of descriptors, creating a storage architecture mismatch with the original on-chain design. The router is also accurate as a classifier but almost neutral as an end-to-end detector component. The experiments show that this is not merely because the classifier is imperfect: the voting mechanics and the interaction between transformations prevent the router's predictions from changing enough decisions.

The main contributions of the project are therefore:

1. a validated reproduction of the original perceptual-hash pipeline;
2. a concrete demonstration that the reproduced **sHash** distance is incompatible with sound BK-tree pruning;
3. a corrected ORB baseline and a controlled pairwise comparison showing a substantial accuracy advantage over **sHash**;
4. a measurement of the accuracy-versus-storage tradeoff introduced by ORB;
5. a query-only reliability router and a measured per-manipulation signal-reliability map;
6. end-to-end Phase D and Phase E experiments showing that routing adds approximately zero incremental gain, together with an experimentally supported explanation of why.

2 Background and Previous Work

2.1 Perceptual image hashes

A perceptual hash compresses visual content rather than exact file bytes. For fixed-width binary hashes, similarity can be measured using Hamming distance,

$$d_H(x, y) = \sum_{i=1}^n \mathbf{1}[x_i \neq y_i].$$

Smaller distance means greater similarity.

aHash reduces an image to a small grayscale representation and thresholds each value relative to the average intensity, producing a coarse description of global luminance structure. **pHash** uses a resized image and a low-frequency discrete cosine transform (DCT), making it more tolerant of some high-frequency detail loss. **hsvHash**, as implemented by the reference **imagehash** library used in our reproduction, is a compact global color-histogram representation. **sHash** is different: it segments an image into salient regions, hashes each segment locally, and produces a variable-length list of segment hashes rather than one fixed binary word.

The original detector combines these four signals with the 2-Minimal rule:

$$\text{duplicate}(q, r) = \mathbf{1} \left[\sum_{i=1}^4 \mathbf{1}[d_i(q, r) \leq t_i] \geq 2 \right].$$

This cross-corroboration is valuable because different transforms damage different representations.

2.2 BK-trees and metric search

A BK-tree stores one object per node and labels child edges by distance from the parent. During a radius search with query distance d and tolerance t , metric inequalities justify exploring only child edges in $[d - t, d + t]$ [3]. Correct pruning therefore depends on a distance that behaves as a metric, in particular on symmetry and the triangle inequality. Hamming distance satisfies these requirements. A variable-length set distance does not automatically do so.

2.3 ORB local feature matching

ORB (Oriented FAST and Rotated BRIEF) detects local keypoints and represents their neighborhoods with binary descriptors [4]. In our pipeline, descriptors are matched with Hamming distance and geometrically verified with a RANSAC-estimated homography. RANSAC separates mutually consistent correspondences from accidental matches [5]. The number of inliers becomes the ORB similarity score. Unlike a global hash, ORB can preserve evidence when an image is cropped, translated, scaled, or rotated, as long as sufficient local structure survives.

A key architectural distinction follows. The original hashes produce compact image-level representations. ORB produces a *set* of binary descriptors. A set of local descriptors therefore calls for descriptor-oriented retrieval such as locality-sensitive hashing (LSH) rather than forcing the entire representation into one BK-tree object [6]. Our accuracy evaluation is pairwise; large-scale retrieval is treated as a separate deployment problem.

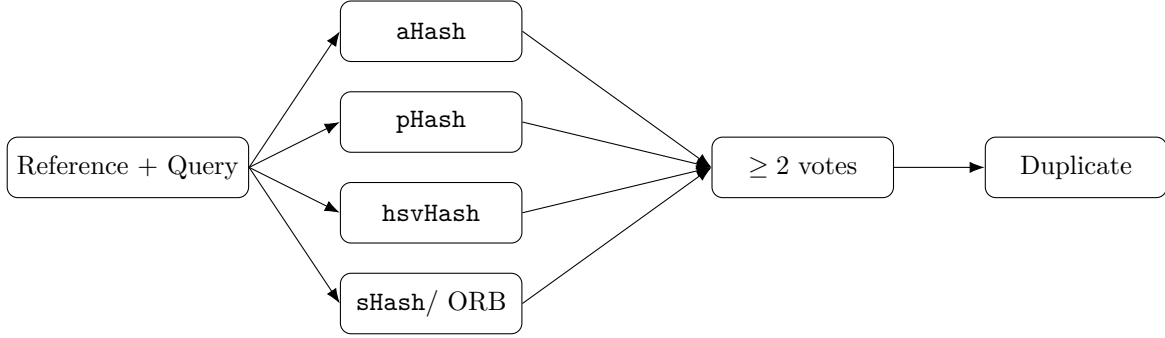


Figure 1: Four-signal detector structure. The original panel uses **sHash**; our primary replacement substitutes ORB while keeping the two-vote rule for controlled comparison.

3 Project Goals and Initial Plan

The initial objective was to preserve the original detector’s answer while reducing unnecessary work. The proposed confidence-gated router would inspect the query, estimate which signals were likely to matter, and avoid expensive searches whose outcome was unlikely to change the final vote. This made the router primarily an efficiency mechanism.

The reproduction phase changed that plan. The **sHash**distance was found to violate BK-tree assumptions, which undermined the premise of optimizing four sound metric indexes. In parallel, benchmarking a new C11 implementation directly against the original Python pipeline would have mixed language-level effects with algorithmic effects. The project therefore pivoted to a Python-only accuracy study: retain the validated C11 hash suite as reproduction groundwork, replace **sHash** with a structurally stronger signal, and re-aim the router from speed toward signal reliability.

The revised research questions were: (1) whether **sHash** can be soundly BK-tree indexed; (2) whether ORB outperforms **sHash** at crop and geometric robustness; (3) whether query-only reliability prediction can improve the complete detector; and (4) what deployment cost the stronger representation introduces.

4 Dataset and Evaluation Methodology

4.1 Image sources and generated labels

The main public corpus contains 3,000 originals: 1,000 Azuki, 1,000 Bored Ape Yacht Club (BAYC), and 1,000 CryptoPunks. The first two collections use conventional artwork resolutions; CryptoPunks are natively 24×24 pixel art. A separate 10,000-image CryptoPunks collection was retained for auxiliary testing rather than merged into the balanced main corpus.

Because the raw images do not carry controlled transformation labels, we generated labeled copies in eight categories: exact copy, resize/crop/reposition, flip/rotate/mirror, pixelated, color modification/saturation, background-color change, text/logo/emoji, and non-duplicate. The category named **flip_rotate_mirror** always includes a non-zero rotation in our generator; it must therefore not be interpreted as a pure-mirror benchmark.

We also used a private reference set shared by the original authors: 405 manipulated CryptoPunk images and a 1,802-row metadata CSV containing original/copy pairs, labels, manipulations, and reference distances. It is used for differential reproduction checks and cross-generator evaluation

but is not redistributed.

4.2 Train/test separation

Base images are split before any manipulation is generated, so variants of the same NFT never appear in both train and test. The single-manipulation corpus contains 2,400 training bases (48,000 generated images / 55,200 pair rows) and 600 test bases (12,000 generated images / 13,800 pair rows). We use an 80/20 split because the router trains a Random Forest over 93 features and eight labels, whereas the original paper primarily tuned a small set of scalar thresholds.

4.3 Pairwise evaluation

Accuracy is evaluated pairwise: each row provides a reference image r , a query q , and a binary label indicating whether q is a manipulated copy of r . Pairwise evaluation matches the dataset schema and prevents gallery membership from influencing the similarity score. Retrieval and indexing remain necessary in deployment, but they are deliberately removed from the accuracy path.

4.4 Metrics and class balance

We report precision, recall, F1, and false-positive behavior. With the usual definitions,

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = 2 \frac{PR}{P + R}.$$

The single-manipulation detector set is 82.6% positive (11,400 positives, 2,400 negatives), so an always-positive classifier already obtains approximately 90.5% F1. The multi-manipulation set is 88.2% positive and gives an always-positive F1 of approximately 93.8%. Raw F1 is therefore never interpreted without precision, recall, and a trivial baseline.

Thresholds are selected on training data and then frozen for test. Detector-level comparisons use a common $\leq 10\%$ false-positive budget on pristine training negatives: `aHash` ≤ 7 , `pHash` ≤ 19 , `hsvHash` ≤ 3 , `sHash` ≤ 21 , and ORB inliers > 23 . The ORB threshold > 16 used later in the geometric signal study answers a different question and is not used for the fair four-signal panel comparison.

5 Reproducing the Original Detector

We reimplemented the four paper hashes in C11, plus `dHash` as a prerequisite for `sHash`. Validation included 600 distinct images spanning example fixtures, the authors' reference images, high-resolution NFTs, native CryptoPunks, and synthetic edge cases. For `sHash`, we matched the full ordered list of segment hashes rather than only the final score. The directional `sHash` distance reproduced all 1,802 author-provided reference rows in the original-to-copy direction.

Two implementation details were especially important. First, exact parity with Pillow required reproducing its mixed float/double arithmetic behavior rather than assuming that variables declared as `float` imply all-float evaluation. Second, the paper's prose description of its color method differs from the actual `colorhash()` implementation used to generate its values: the executed implementation is a global color histogram. For reproduction, the executed reference behavior was treated as authoritative.

This stage was not intended as the final contribution. Its role was to establish that later anomalies were properties of the reproduced method rather than consequences of an unrelated reimplementation error.

6 Finding 1: sHash Is Not a Sound BK-Tree Metric

6.1 Directional distance

Let image A contain segment hashes $\{a_1, \dots, a_m\}$ and image B contain $\{b_1, \dots, b_n\}$. The reproduced `sHash` distance is directional:

$$d_s(A, B) = \frac{1}{m} \sum_{i=1}^m \min_j d_H(a_i, b_j).$$

Only source-side segments are required to find a match. Extra target segments can therefore be free in one direction and costly in the reverse direction.

For a concrete example, let P_1 contain three segments and P_2 contain those same three plus one unmatched segment. If the three good matches have distance 2 and the unmatched segment’s nearest neighbor is distance 34, then

$$d_s(P_1, P_2) = 2, \quad d_s(P_2, P_1) = 10.$$

The distance is not symmetric. The authors’ reference CSV demonstrates the same phenomenon empirically: original-to-copy distances reproduce all 1,802 rows, while the reverse direction reproduces only 240.

6.2 Triangle inequality

A second counterexample uses $A = \{\text{cat}\}$, $B = \{\text{cat}, \text{dog}\}$, and $C = \{\text{dog}\}$. With same-object distances near 2 and cat-to-dog distance near 30,

$$d(A, B) = 2, \quad d(B, C) = 16, \quad d(A, C) = 30,$$

so $30 \leq 2 + 16$ is false. A composite representation can “bridge” two otherwise distant objects.

6.3 Concrete pruning failure

The consequence is not merely definitional. Suppose a query $q = \{\text{cat}\}$ has a genuine duplicate $X = \{\text{cat}\}$ with $d(q, X) = 2$, but X is stored below a composite node $B = \{\text{cat}, \text{dog}\}$ with stored edge $d(B, X) = 16$. If $d(q, B) = 2$ and search tolerance is $t = 5$, BK-tree pruning explores only child edges in $[-3, 7]$. The edge 16 is pruned even though X is a valid radius-5 neighbor.

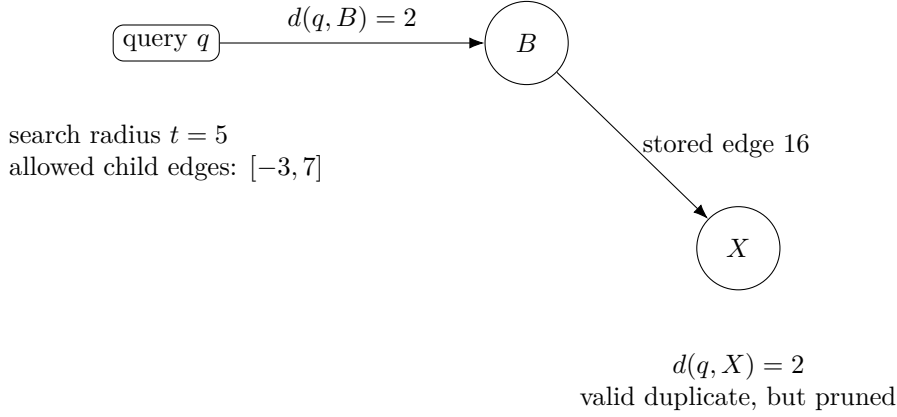


Figure 2: Concrete BK-tree false-negative construction caused by the non-metric `sHash`distance.

We raised the finding with one of the authors. The response indicated that the metric-property issue had not been noticed and might explain imperfect tree behavior, but no formal resolution was provided. We therefore make the narrower claim established directly by the implementation and mathematics: the reproduced `sHash`distance is non-metric, and standard BK-tree pruning can miss a valid neighbor.

7 Project Pivot and Correction of the ORB Baseline

7.1 Why ORB

`sHash` was intended to improve robustness to cropping and partial-image transformations. ORB targets the same broad role using local keypoints and geometrically verified descriptor matches. It therefore became the natural candidate replacement after the `sHash` indexing issue was identified.

7.2 Retiring the old baseline

An earlier ORB retrieval pipeline reported F1 73.6%, precision 82.7%, recall 66.3%, and accuracy 60.7%. Before building on it, we inspected the gallery and found only 2,000 of the expected 3,000 originals. Every CryptoPunk was missing.

The cause was resolution. Native CryptoPunks are 24×24 , while ORB’s default patch size and edge threshold are 31 pixels. Descriptor extraction returned no usable descriptors, and gallery construction silently skipped those images. The resulting confusion matrix was dominated by collection membership. A verification script that performed no image matching at all - only checking whether the relevant filename was present in the gallery - reproduced all four confusion-matrix cells exactly: 30,210 TP, 15,390 FN, 6,306 FP, and 3,294 TN. The ORB threshold sweep was almost flat, which further confirmed that the similarity score was not controlling most decisions.

The corrected evaluation introduced two changes. First, images are normalized to a common ORB working resolution, using nearest-neighbor upscaling for pixel art. Native CryptoPunks move from zero keypoints to hundreds of usable keypoints, although upscaling cannot recreate information absent from the original. Second, ORB accuracy is evaluated pairwise so gallery construction cannot manufacture the result.

8 Finding 2: ORB Outperforms sHash at Its Intended Role

The direct comparison uses 3,600 positive geometric pairs (crop/reposition and rotation-dominated examples) plus 2,400 negatives. ORB uses a threshold selected honestly on train (> 16 RANSAC inliers). **sHash** is deliberately given an oracle advantage: its best threshold is swept on the test set itself (≤ 28).

Table 1: Direct geometric-subset comparison.

Signal	Threshold	Precision	Recall	F1
sHash (oracle-best)	$d \leq 28$	66.5%	89.8%	76.4%
ORB (train-tuned)	inliers > 16	91.2%	90.0%	90.6%

The superficially similar recall values are misleading. At its oracle threshold, **sHash** flags 1,630 of 2,400 negatives. At ORB’s 91.2% precision, **sHash** reaches only 27.2% recall, compared with ORB’s 90.0%.

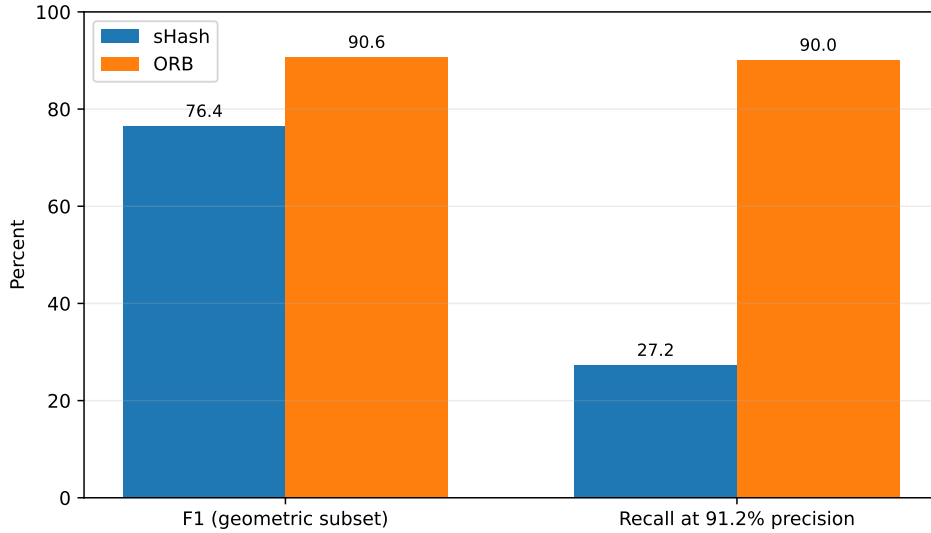


Figure 3: ORB vs. **sHash** on the geometric subset. **sHash** receives an oracle threshold; ORB still dominates both F1 and recall at equal precision.

Per-category evaluation refined the interpretation: ORB is better described as a *structure specialist* than a narrowly geometric specialist. At the signal-level threshold it detects 100% of exact copies, 98.3% of crop/reposition cases, 100% of text/logo/emoji overlays, 98.9% of background-color changes, 79.8% of color modifications, but only 28.5% of pixelated samples. The common factor is preservation of local structure.

Low-resolution CryptoPunks remain the main limitation. At a clean 9.9% false-positive operating point, ORB detects 100% of geometric copies for Azuki and BAYC but 68.5% for CryptoPunks. A diagnostic split of the `flip_rotate_mirror` category also showed that pure mirroring is not the weakness: pure mirrors were detected at 100%, while rotations of low-resolution CryptoPunks caused most misses.

9 Finding 3: ORB’s Deployment Cost

The original detector’s compact hashes require only tens of bytes per image. ORB’s accuracy comes from preserving many local descriptors. At the default setting, a normalized image produces roughly 455 descriptors; at 32 bytes per descriptor, this is about 14,560 bytes per image.

Table 2: Measured representation size.

Representation	Approx. bytes/image	Relative to 400-byte reference transaction
aHash+ pHash+ hsvHash	24	$0.06\times$
sHash(median)	32	$0.08\times$
ORB, default working point	14,560	$36.4\times$

We therefore measured an accuracy-versus-bytes curve rather than treating the OpenCV default `nfeatures=500` as meaningful. Around 128 requested features (about 122 actual descriptors), payload falls to 3,888 bytes and F1 remains 87.5% on the geometric subset. At 32 descriptors, storage is 1,024 bytes but F1 falls to 76.1%, approximately the oracle-best sHashlevel.

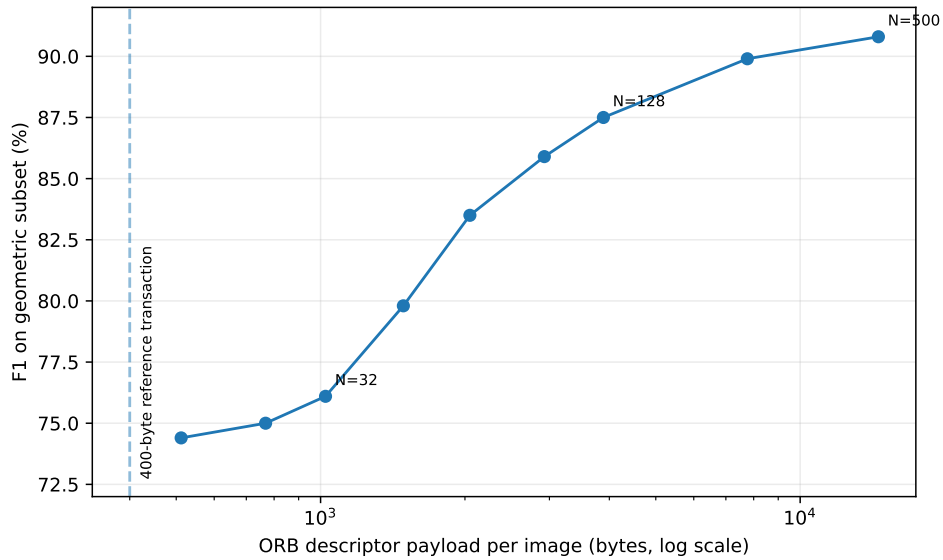


Figure 4: ORB accuracy versus descriptor payload. Reducing descriptor count lowers storage substantially, but the compact-storage regime removes most of the measured accuracy advantage.

This is an architectural tradeoff rather than an ordinary threshold issue. Possible alternatives include storing descriptors off-chain with an on-chain commitment, recomputing descriptors from the asset when needed, or aggregating descriptors into a fixed-size vector. Each option forfeits some property of the original design or of ORB’s geometrically verified matching.

10 Manipulation-Aware Router

10.1 Query-only constraint

The router receives only the query image. It cannot compare against the original to infer what changed. All 93 features are therefore absolute single-image descriptors. A Random Forest [7]

maps them to probabilities over the eight manipulation labels. The detector consumes reliability decisions rather than the hard class label:

$$P(\text{detail broken}) = P(\text{pixelated}),$$

$$P(\text{color changed}) = P(\text{color modification}) + P(\text{background change}).$$

The first is used to distrust ORB; the second to distrust `hsvHash`.

The exact-copy and non-duplicate labels illustrate why raw eight-class accuracy is not the primary objective. From the query pixels alone, a byte-identical exact copy is visually indistinguishable from a pristine image. What matters is whether the resulting prediction implies a safe reliability action.

10.2 Reliability results

Raw multiclass accuracy is 89.2%. The task-relevant reliability decisions are stronger: the detail-broken decision reaches 100% precision, recall, and F1 on the generated test distribution, and the color-change decision reaches 98.1% precision, 97.0% recall, and 97.5% F1.

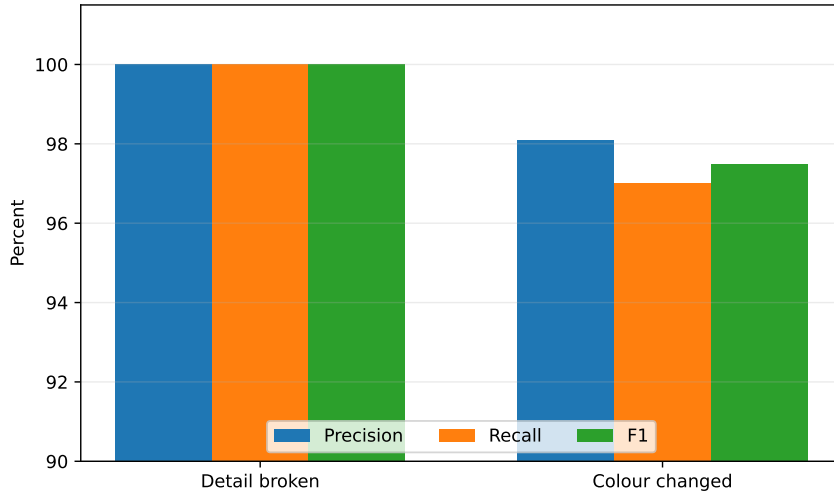


Figure 5: Router reliability decisions on the held-out generated test split.

Important features include unique-color ratio, flat-neighbor fraction, transparency/corner statistics, Laplacian variance, and histogram-combing measures. The first two are strong pixelation cues; transparency can reveal our generator’s rotated canvases; histogram combing can reveal some color edits.

10.3 Measured signal-reliability map

Rather than assume which signal each manipulation breaks, we measured detection and false-positive behavior at train-derived operating points. Several results contradicted intuition. Pixelation collapses ORB to 28.5% detection but `pHash` remains at 65.5% and `hsvHash` at 79.3%; therefore the detail flag gates ORB alone in the ORB panel. Conversely, `hsvHash` falls to 16.1% on background-color changes and 28.3% on other color edits. `aHash` and `pHash` fall to 5.0% and 2.8% respectively on the rotation-dominated category.

The most important observation is that broken signals are generally *silent*. Their false-

positive rates often fall together with their true-positive rates. For example, `aHash` detects only 5.0% of rotation-dominated positives and misfires on only 0.4% of corresponding wrong-original comparisons. A silent signal already behaves like an abstention; explicitly dropping it does not change a two-vote decision.

10.4 Cross-generator limitation

The authors’ independent CryptoPunk set exposes domain shift. Their images are pre-upscaled RGB, while ours are native-resolution RGBA before processing. Transparent-corner and histogram-combing cues disappear, and the router over-predicts pixelation. The color-reliability decision degrades severely and the geometry trigger has 0% recall on the relevant external cases. The router’s strong reliability numbers must therefore be interpreted as within-distribution results, not as evidence of universal manipulation recognition.

11 Phase D: Does Routing Improve the Detector?

Phase D separates the ORB replacement from the incremental value of routing. Three configurations use the same pairwise test split and common false-positive operating points:

- A: static {`aHash`, `pHash`, `hsvHash`, `sHash`}, the paper-style panel;
- B: static {`aHash`, `pHash`, `hsvHash`, ORB}, isolating the swap;
- C: routed ORB panel, isolating routing on top of the swap.

Table 3: Phase D single-manipulation detector comparison.

Detector	Precision	Recall	F1
A: static + <code>sHash</code>	97.6%	70.3%	81.7%
B: static + ORB	97.8%	79.2%	87.5%
C: routed + ORB	97.8%	79.2%	87.5%

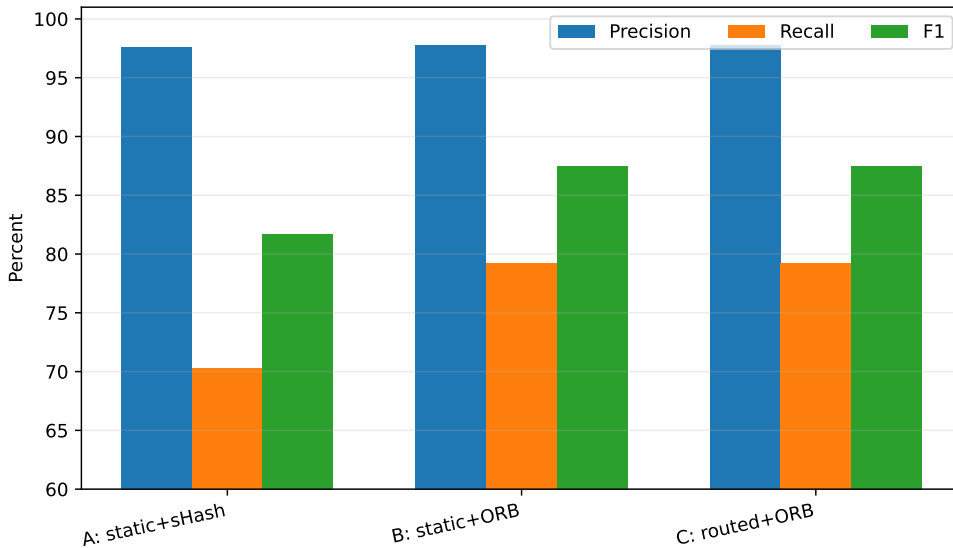


Figure 6: Phase D result: the ORB replacement accounts for the measurable gain; routing adds approximately zero.

The gain is located where expected. Rotation-dominated detection rises from 14.1% to 58.4%, crop/reposition from 69.9% to 80.9%, while pixelation falls modestly from 71.1% to 66.4%. The ORB panel therefore trades a small pixelation loss for a much larger geometric gain.

Why does the router add nothing? With four or three trusted signals, the healthy-panel rule still requires two votes. A useful routing action appears only when two signals are distrusted simultaneously, allowing the quorum to fall to one-of-two. The single-manipulation generator almost never creates that state.

A simulation of a geometry reliability flag demonstrates that the lever is real. Distrusting both `aHash` and `pHash` rotation reduces the healthy panel to `{hsvHash, ORB}` and allows one-of-two; flip/rotation recall rises from 58.4% to 97.7% with no measured FP increase. But the trigger itself is generator-specific and scores 0% cross-generator, so the gain is not used as a headline result.

11.1 Voting quorum and prevalence

Sweeping the static quorum on the positive-heavy test set makes $k = 1$ appear best. This is a base-rate artifact. Reweighting to lower assumed copymint prevalence reverses the conclusion: at 2%, 10%, and 25% prevalence, $k = 2$ outperforms $k = 1$.

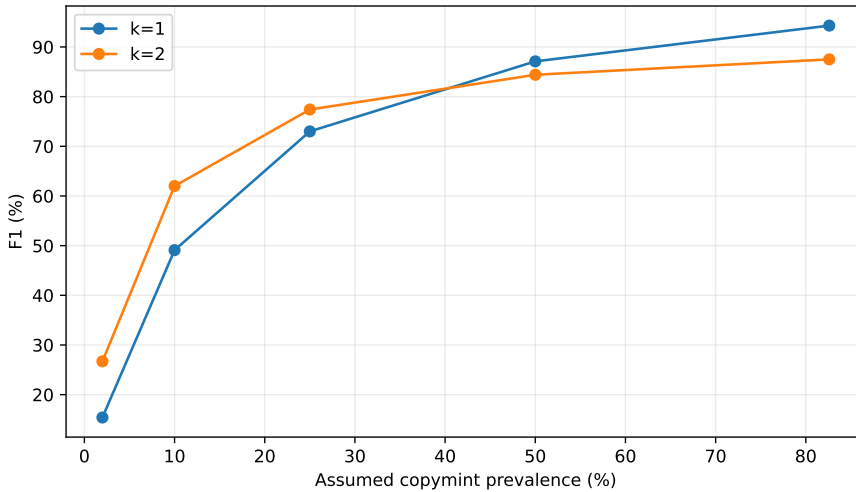


Figure 7: The preferred quorum depends strongly on prevalence. The permissive $k = 1$ rule is favored only by the unusually positive-heavy generated evaluation set.

We also compare against the paper’s frozen published thresholds rather than only our re-tuned baseline. On our test set, the frozen paper configuration yields $P=98.1\%$, $R=64.1\%$, $F1=77.5\%$; replacing only `sHash` with ORB gives $P=98.1\%$, $R=77.2\%$, $F1=86.4\%$. Thus the swap improves F1 by 8.9 points against the as-published configuration and 5.8 points against the stronger re-tuned baseline.

12 Phase E: Multi-Manipulation Stress Test

Phase D suggested that routing needs two simultaneous reliability failures. Phase E constructs that missing state directly by composing two transformations. The six non-exact manipulation families are combined as all 15 unordered pairs in both orders, giving 30 composite categories. The multi-test set contains 18,000 positive pairs (600 per ordered pair) and 2,400 negatives.

The storage-compatible hypothesis uses the original compact panel and asks whether routing can manage **sHash** rather than replacing it with the large ORB representation. Adding **sHash** to the reliability map shows that pixelation reduces standalone **sHash** detection to 38.1% while leaving false-positive behavior close to its pristine baseline. The detail flag can therefore reasonably distrust **sHash** under pixelation.

A useful two-flag state would combine pixelation (distrust **sHash**) with a color change (distrust **hsvHash**), leaving **{aHash, pHash}** and enabling one-of-two voting. However, the single-label router’s color flag fires 0% whenever pixelation is present. To distinguish classifier competition from missing evidence, we trained independent binary heads for detail and color. The result is the same: the color head fires 76–100% on color combined with non-pixelation transformations but 0% on pixelation+color in both orders. Under the tested severe pixelation process, the color forensic cue used by this router is erased.

Table 4: Phase E multi-manipulation comparison.

Detector	Precision	Recall	F1	$\Delta F1$ vs A
A: static + sHash	96.7%	36.4%	52.9%	–
B: routed, same compact panel	96.7%	36.4%	52.9%	+0.01
B’: static + ORB swap	97.3%	45.1%	61.6%	+8.7

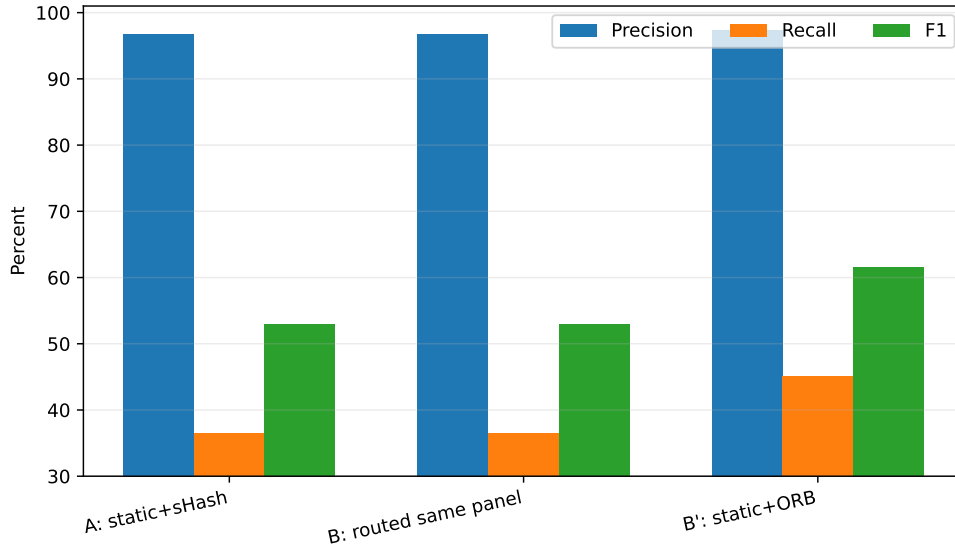


Figure 8: Phase E result on two-manipulation composites. Routing remains neutral; the ORB replacement continues to improve recall at similar precision.

Routing without a static fallback actually regresses (F1 50.3%) because it can remove a sometimes-useful **sHash** vote without reaching the second reliability condition needed to relax the quorum. The fallback prevents harm but also leaves the router with essentially no measurable benefit.

The strongest defensible interpretation is scoped to the tested manipulation model: the two-distrust state required by this router is *forensically self-cancelling*. Pixelation is the relevant detail-breaker, and in our generated regime it removes the color cue needed to observe the simultaneous color failure. This is not a universal impossibility result for all future routers or milder pixelation settings.

13 Discussion

13.1 Correctness before optimization

The `sHash` result shows why a detector representation and its index cannot be designed independently. A threshold can be statistically tuned while the retrieval structure remains mathematically unsound. The first priority is therefore preserving candidate-search correctness; only then does optimizing runtime make sense.

13.2 Representation quality versus deployment cost

ORB demonstrates the opposite tension. It substantially improves the structure-aware similarity signal, yet the richer representation no longer fits the metadata assumptions of the original system. An algorithmic improvement can therefore be real while still requiring a system-level redesign. The appropriate conclusion is not “ORB is deployable as-is,” but “ORB establishes the value of stronger local structure and quantifies what that value costs.”

13.3 Component accuracy versus end-to-end utility

The router is a useful counterexample to evaluating modules in isolation. Within the generated distribution, reliability classification is excellent. End-to-end gain remains approximately zero because the detector’s voting mechanics do not change when one already-silent signal is removed. The downstream action, not only the classifier’s accuracy, determines utility.

13.4 Compound transformations are interactive

Phase E shows that transformation evidence is not necessarily additive. One operation can erase the forensic trace of another. A robust multi-manipulation detector should therefore model reliability properties directly rather than assuming that independently recognizable manipulation labels remain visible after composition.

13.5 Evaluation design as technical work

Several apparently strong results were invalidated or corrected only because the evaluation was treated as part of the engineering problem. The missing-gallery ORB baseline, the always-positive `sHash` threshold, the unequal ORB false-positive operating point, and the merged rotation/mirror label all demonstrate that a plausible metric can still answer the wrong question. Sanity checks against class balance, parameter sensitivity, and dataset composition were essential to the final conclusions.

14 Difficulties and Lessons Learned

The project repeatedly required reproducing unexpected behavior rather than immediately optimizing it. Exact hash parity depended on implementation-level arithmetic and library semantics. ORB imposed a minimum practical input resolution that the global hashes did not, and its failure mode was silent. Dataset labels combined transformations of very different difficulty. Threshold selection on a positive-heavy corpus produced superficially attractive but trivial solutions.

Several recurring lessons emerged:

1. **Reproduce executed software, not only prose.** Defaults, preprocessing, numeric precision, and library behavior are part of the experiment.
2. **Validate corpus invariants early.** Counts per collection, class balance, and descriptor coverage should be asserted before trusting a metric.
3. **Treat extreme sweep results as warnings.** A threshold at a grid boundary often indicates a parsing, class-balance, or objective problem.
4. **Measure signal failure modes.** A broken signal may become silent rather than noisy, which changes the value of routing completely.
5. **Explain null results mechanistically.** The router’s near-zero contribution became informative only after experiments identified the quorum and cue-erasure mechanisms behind it.

15 Limitations and Future Work

15.1 Generator-specific router cues

Most manipulations are PIL-generated. The cross-generator test shows that transparency, histogram-combing, and resolution conventions can become shortcuts. Future work should train across multiple independent generators, interpolation filters, image formats, resolutions, screenshots, recompression pipelines, and manually edited examples.

15.2 Pixelation severity

The pixelation process downsamples images to roughly 4–12% of their original dimensions before resizing them back. This is severe. The detail-classification and cue-erasure conclusions should therefore be repeated across a controlled severity curve to determine whether milder pixelation leaves a region where both reliability flags remain observable.

15.3 Direct reliability prediction

The current router predicts manipulation classes and then maps them to signal trust. A more direct formulation would predict a vector such as

$$[r_a, r_p, r_{hsv}, r_{orb}],$$

where each component estimates the reliability of a detector signal. This would align training targets with the downstream decision and reduce dependence on whether a semantic manipulation label is recognizable.

15.4 Large-scale retrieval

Pairwise evaluation establishes signal quality but not full search-system performance. A production ORB path would require scalable descriptor retrieval, candidate generation, and geometric verification. Future work should benchmark retrieval recall, candidate count, latency, memory, index-build cost, and the interaction between retrieval errors and final detection.

15.5 Compact structure-aware representations

The descriptor-budget curve suggests that simply truncating ORB eventually removes its advantage. A more promising direction is a representation designed from the start to preserve partial/geometric similarity in fewer bytes: compact learned binary descriptors, compressed local-feature sets, or carefully evaluated aggregation schemes such as Bag-of-Visual-Words or VLAD-like encodings. Any alternative should be judged jointly on detection accuracy, storage size, searchability, and preservation of geometric verification.

15.6 Low-resolution image regimes

CryptoPunks remain difficult because upscaling creates spatial extent but not new information. A future detector could use resolution-aware routing: detect the native image regime first and select a representation suited to pixel art rather than forcing one local-feature method to dominate every collection.

15.7 Public cross-generator benchmark

The private author reference set is valuable but small and CryptoPunk-focused. A stronger benchmark should contain multiple collections, independently generated single and compound manipulations, manipulated negatives, separate rotation and mirror labels, multiple source resolutions, and permission for public redistribution.

16 Conclusion

This project began as an extension of a four-hash NFT duplicate detector and evolved into an investigation of representation correctness, geometric robustness, deployment cost, and adaptive voting. The reproduction phase established a trustworthy baseline and exposed the first central finding: the reproduced `sHash`distance is asymmetric and violates the triangle inequality, so it does not satisfy the metric assumptions required for sound BK-tree pruning.

Replacing `sHash`with ORB produced the clearest positive result. On the geometric subset, ORB achieved 90.6% F1 compared with an oracle-best 76.4% for `sHash`, and at equal precision its recall was 90.0% versus 27.2%. In the complete detector, the replacement improved F1 from 81.7% to 87.5% at a common false-positive operating point. On the harder two-manipulation benchmark, the replacement still improved F1 by approximately 8.7 points. The gain is therefore not an artifact of one isolated comparison.

The stronger representation also revealed a deployment boundary. ORB emits hundreds of descriptors and requires orders of magnitude more metadata than the original compact hashes. It cannot simply be substituted into the original on-chain storage model without changing where features are stored or computed.

The routing experiments produced the main negative result. The router predicts the tested reliability states accurately within the generated distribution, yet adds approximately zero end-to-end gain. Broken signals usually become silent rather than noisy, so removing a single one does not change a two-vote quorum. Phase E then constructed two-manipulation examples intended to trigger quorum relaxation, but severe pixelation also erased the color forensic cue required to observe the second failure. Under the tested features and transformation regime, the useful two-distrust state is therefore difficult to reach in a detectable form.

The project consequently contributes both a practical improvement and a boundary on a promising idea. The measurable improvement comes from changing the image representation itself: **sHash**→ORB. The router is valuable primarily as an experimental instrument that exposes how signal failures and voting rules interact, rather than as a source of additional detector accuracy in the current system. More broadly, the work shows that robust copymint detection must consider the representation, distance mathematics, index structure, storage model, dataset construction, transformation interactions, and final decision rule together.

A Key Experimental Operating Points

Table 5: Thresholds used for the fair detector panels.

Signal	Duplicate condition
aHash	Hamming distance ≤ 7
pHash	Hamming distance ≤ 19
hsvHash	Hamming distance ≤ 3
sHash	directional distance ≤ 21
ORB (detector panels)	RANSAC inliers > 23
ORB (geometric signal study)	RANSAC inliers > 16

Table 6: Main headline results.

Experiment	Comparison	Result
Geometric signal study	ORB vs. oracle sHash	90.6 vs. 76.4 F1
Phase D, equal-FP	static sHash vs. static ORB	81.7 $\rightarrow$ 87.5 F1
Phase D, routing	static ORB vs. routed ORB	87.5 $\rightarrow$ 87.5 F1
Phase E, multi-manip	static sHash vs. static ORB	52.9 $\rightarrow$ 61.6 F1
Phase E, routing	static compact vs. routed compact	52.9 $\rightarrow$ 52.9 F1

B Reproducibility Commands

The project repository contains cached scores and scripts for reproducing the main measurements. Representative commands are:

```
python/geometric/compare_shash.py --split test
python/geometric/descriptor_budget.py --split test --budgets 500,128
python/router/evaluate_router.py --split test
python/detector/evaluate_routed.py --split test
python/detector/evaluate_multi.py
python/detector/evaluate_multi.py --single
python/detector/evaluate_reference.py
```

On a fresh checkout, the generated datasets and cached signal CSVs must be regenerated before these evaluation scripts can run.

C Data Notes

The main generated dataset uses the public Kaggle collections [shaunmak/nft-classifier](#) and the separate [tunguz/cryptopunks](#) corpus for auxiliary testing. The private manipulation reference set supplied by the original authors is intentionally not redistributed. Base-image splitting is performed before manipulation generation to prevent variants of one original from crossing the train/test boundary.

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